Solving Logarithmic Equations with Domain Checks Answer Key

Precalculus / Advanced Algebra

Quick Answers

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1. 32
2. 13
3. 8
4. $\frac{\log(x)}{\log(2)} + 3$
5. 4
6. -4
7. $1 + e^2$
8. 2, -6
9. 10.54
10. $\frac{13}{9}$
11. 20.99
12. 3

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1. $\log_2 x = 5$. *Domain:* $x > 0$. *Rewrite:* $\log_b A = c \Leftrightarrow A = b^c$, so $x = 2^5$.

$$x = 32$$

Check: $32 > 0$, and $\log_2 32 = 5$ ✓. $\boxed{x = 32}$

2. $\log_3(x-4) = 2$. *Domain:* the argument must be positive, $x - 4 > 0$, so $x > 4$. *Rewrite:* $x - 4 = 3^2 = 9$. *Solve:* $x = 13$. *Check:* $13 > 4$ ✓, and $\log_3(13-4) = \log_3 9 = 2$ ✓. $\boxed{x = 13}$

3. $\log_5(2x+1) = \log_5(x+9)$. *Domain:* $2x + 1 > 0$ and $x + 9 > 0$, so $x > -\frac{1}{2}$ (the tighter of the two). *Solve:* logarithms with the same base are equal only when their arguments are equal, so

$$2x + 1 = x + 9 \quad \Longrightarrow \quad x = 8$$

Check: $8 > -\frac{1}{2}$ ✓; both sides become $\log_5 17$ ✓. $\boxed{x = 8}$

4. Rewrite $\log_2(8x)$. Product rule: $\log_b(MN) = \log_b M + \log_b N$, so

$$\log_2(8x) = \log_2 8 + \log_2 x$$

and $\log_2 8 = 3$ because $2^3 = 8$. $\boxed{\log_2(8x) = 3 + \log_2 x}$

Valid for $x > 0$: the rewriting splits one logarithm into two, and the new $\log_2 x$ needs its own positive argument. This is the move that turns $\log_2(8x) = k$ into a one-logarithm equation.

5. $\log_2 x + \log_2(x-2) = 3$. *Domain:* $x > 0$ *and* $x - 2 > 0$, so $x > 2$. *Combine:* product rule gives $\log_2\big(x(x-2)\big) = 3$. *Rewrite and solve:*

$$x(x-2) = 2^3 = 8$$
$$x^2 - 2x - 8 = 0$$
$$(x-4)(x+2) = 0$$

Candidates $x = 4$ and $x = -2$. Only 4 lies in $x > 2$. *Check:* $\log_2 4 + \log_2 2 = 2 + 1 = 3$ ✓. $\boxed{x = 4}$

6. **Why $x = -2$ fails.** Evaluate the argument of the second logarithm at the candidate:

$$x - 2 = (-2) - 2 = -4$$

$$\boxed{-4}$$

The candidate would require $\log_2(-4)$, and a logarithm is undefined for a negative argument, so $x = -2$ is extraneous — it solves the quadratic that came *after* the logarithms were cleared, not the original equation. The domain of the original equation is $x > 2$ (both $x > 0$ and $x - 2 > 0$ must hold). *Note for the grader: the one-sentence explanation is judged by you; the evaluated argument -4 is machine-verified.*

7. $\ln(x - 1) = 2$. *Domain:* $x - 1 > 0$, so $x > 1$. *Rewrite* with base e: $x - 1 = e^2$.

$$x = e^2 + 1$$

Check: $e^2 \approx 7.389$, so $x \approx 8.389 > 1$ ✓. $\boxed{x = e^2 + 1}$

8. $\log_2(x^2 + 4x) = \log_2 12$. *Domain:* $x^2 + 4x > 0$, i.e. $x(x+4) > 0$, which holds for $x < -4$ or $x > 0$. *Solve:* equal logarithms, equal arguments:

$$\begin{aligned} x^2 + 4x &= 12 \\ x^2 + 4x - 12 &= 0 \\ (x - 2)(x + 6) &= 0 \end{aligned}$$

Candidates $x = 2$ and $x = -6$. *Check both:* at $x = 2$, the argument is $4 + 8 = 12 > 0$ ✓; at $x = -6$, the argument is $36 - 24 = 12 > 0$ ✓ (and $-6 < -4$, inside the domain).

$$\boxed{x = 2 \text{ or } x = -6}$$

A negative candidate is *not* automatically extraneous — what matters is the sign of the argument, not the sign of x.

9. $\ln(2x - 1) = 3$, **to the nearest hundredth.** *Domain:* $2x - 1 > 0$, so $x > \frac{1}{2}$. *Rewrite:* $2x - 1 = e^3$, so

$$x = \frac{e^3 + 1}{2} \approx \frac{20.0855 + 1}{2} = 10.5428\ldots$$

Check: $10.54 > \frac{1}{2}$ ✓. $\boxed{x \approx 10.54}$

Keep the exact form until the last line: rounding e^3 early moves the answer.

10. $\log(x+3) - \log(x-1) = 1$ **(base 10).** *Domain:* $x+3>0$ and $x-1>0$, so $x>1$. *Combine* with the quotient rule, then rewrite:

$$\begin{aligned}
\log\left(\frac{x+3}{x-1}\right) &= 1 \\
\frac{x+3}{x-1} &= 10^1 = 10 \\
x+3 &= 10x - 10 \\
13 &= 9x
\end{aligned}$$

$\boxed{x = \frac{13}{9}}$

Check: $\frac{13}{9} \approx 1.44 > 1$ ✓, and the arguments are $\frac{40}{9}$ and $\frac{4}{9}$, whose quotient is 10 ✓.

11. $\log_2(x+5) = 4.7$**, to the nearest hundredth.** *Domain:* $x+5>0$, so $x>-5$. *Rewrite:* $x+5 = 2^{4.7}$, so

$$x = 2^{4.7} - 5 \approx 25.9921 - 5 = 20.9921$$

Check: $20.99 > -5$ ✓. $\boxed{x \approx 20.99}$
A non-integer exponent is fine — exponential form does not require b^c to be a whole number.

12. **Synthesis:** $2\log_5 x - \log_5(x+6) = 0$**.** *Domain:* $x>0$ and $x+6>0$; together, $x>0$. *Combine* with the power rule then the quotient rule:

$$\begin{aligned}
\log_5 x^2 - \log_5(x+6) &= 0 \\
\log_5\left(\frac{x^2}{x+6}\right) &= 0 \\
\frac{x^2}{x+6} &= 5^0 = 1 \\
x^2 - x - 6 &= 0 \\
(x-3)(x+2) &= 0
\end{aligned}$$

Candidates $x=3$ and $x=-2$; the domain $x>0$ eliminates -2, which would require $\log_5(-2)$. *Check:* $2\log_5 3 - \log_5 9 = \log_5 9 - \log_5 9 = 0$ ✓. $\boxed{x = 3}$